Aman Malik

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Profile

Computer Science undergraduate with hands-on experience in machine learning, deep learning, computer vision, NLP, and agentic AI systems. Experienced in developing production-grade AI agents, RAG applications, and intelligent systems from scratch, with a strong interests in AI research and real-world intelligent systems.

Education

Bachelor of Science in Computer Science 08/2023 – Present
FAST NUCES Karachi, Pakistan
CGPA: 3.73

Technical Skills

Languages
Python, C/C++, SQL, PostgreSQL

AI/ML & Frameworks
PyTorch, TensorFlow/Keras, Scikit-learn, Hugging Face Transformers, LangChain, OpenCV, NumPy, Matplotlib

Generative AI & Agentic Systems
LLMs, RAG, Vector Databases, Agentic AI, Tool Calling, Multi-Agent Systems, n8n

Domains
Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Conversational AI, Recommender Systems

Professional Experience

Agentic AI Engineer 04/2026 – Present
BizBuddy Karachi, Pakistan
- Built and improved production-grade agentic AI systems for real-world business workflows using LLMs, n8n, APIs, and database integrations.
- Independently developed AI agents from scratch, including tool-calling, workflow logic, and reliability safeguards.
- Debugged and optimized existing agents to improve accuracy, robustness, and end-to-end workflow reliability.

Publications

Beyond Prompt Engineering: A Controlled Evaluation of Deterministic Guardrails in a Transactional LLM Agent 2026
Zenodo
- Conducted a controlled evaluation of **prompt-only vs. tool-grounded LLM agents across 20 transactional scenarios**, achieving **70% vs. 90% task success**.
- Analyzed **deterministic guardrails, tool grounding, routing/orchestration failures, and safe transaction validation** in real-world LLM agent workflows.
- Published as a **research preprint on Zenodo** with an accompanying technical article on Medium.

Do Multi-Agent LLM Systems Actually Help? A Controlled Evaluation of Single-Agent, LLM-Orchestrated, and Deterministically Orchestrated Architectures 2026
Zenodo
- Evaluated **three agent** architectures across **30 tasks** and **270 runs**, **measuring success rate, reliability, latency, and token usage**.
- Achieved **93.33%** task success with deterministic orchestration, near the **94.44%** single-agent baseline and above **74.44%** LLM-based orchestration.
- Published as a **research preprint on Zenodo**, with an accompanying technical article on Medium.

Projects

ResumeLens – AI-Powered Resume Analyzer
Python | FastAPI | React | Pydantic | NLP
- Built a full-stack AI application using React, FastAPI, and Gemini 2.5 Flash to evaluate resumes against job descriptions, identifying skill gaps, experience mismatches, and ATS keywords.
- Implemented schema-constrained structured outputs with Pydantic validation and an AI career coach for personalized, job-specific resume improvements.

Nexus – Autonomous AI Coding Agent
Python | OpenAI API | FastAPI | Git | pytest
- Built an autonomous coding agent with a VS Code extension, CLI, and API, capable of repository exploration, semantic code search, task planning, and multi-file code modifications.
- Implemented tool-calling workflows with iterative execution, test-driven verification, error-guided repair, and Git checkpoint/rollback to help recover from failed changes while preserving existing user work.
- Developed a task-based benchmark with acceptance checks, timeout handling, and independent test verification to evaluate coding-agent reliability.